Nikhil Ranjan

Rust Systems Engineer | Substrate Runtimes | On-chain Exchange

Berlin, Germany niklabh811@gmail.com niklabh.github.io
github.com/niklabh linkedin.com/in/niklabh x.com/niklabh

Summary

Rust systems engineer with 12+ years shipping production infrastructure. Contributed to the Polkadot and Kusama launches at Parity, including Substrate governance-pallet state transitions. Built stableswap AMM logic for **Vulture Finance** and worked with **ArthSwap** on their Astar AMM. Built **VelociDB** (MVCC, lock-free structures, io_uring, SIMD). CTO of **Polkassembly**.

Experience

Polkassembly *Berlin*
CTO & Co-founder Jul 2023 – Present

- Founded and architected Polkadot's governance platform: 1,700+ referenda tracked, $300M+ in treasury decisions facilitated, 250K+ participants.
- Built Rust and Substrate backends for proposal and account state, validator coordination, delegated voting, and transaction submission.
- Grew the open-source organization from 1 to 20 engineers; led architecture, hiring, and code review.

Astar *Berlin*
Senior Protocol Engineer Feb 2022 – Jun 2023

- Developed Astar's parachain runtime and governance logic in Rust and Substrate.
- Worked on astar.js, astar apps, astar base, and astar portal.
- Worked with the Astar gas estimation API and faucet.
- Worked with ArthSwap on their Astar AMM: constant-product pools, swap execution, and liquidity accounting.

Parity Technologies *Berlin*
Principal Engineer Dec 2019 – Jul 2021

- Core contributor to the Polkadot and Kusama launches: shipped Substrate runtime code in Rust.
- Implemented the governance pallet's state transitions: proposals, referenda, voting, and scheduled enactment.
- Shipped validator lifecycle and staking flows for account-state updates, hardened for adversarial validator behavior.

Earlier — **Paytm** (2013–2015), founding engineer, payments microservices to 100M+ users. **Tokopedia** (2016–2017), Tokopedia Lite, Google's Best PWA. **Frontier Car Group** (2017–2019), marketplaces across Latin America, Africa, and Asia.

Selected Projects

Vulture Finance *Stableswap AMM.*
Built stableswap and AMM logic: swap execution, pool state, and liquidity accounting for correlated assets.

VelociDB *Embedded database engine in Rust.*
SQLite-inspired storage engine for modern hardware: MVCC, async I/O on io_uring, lock-free structures, and SIMD vectorization.

Oxide *Binary-native browser in Rust.*
WebAssembly sandbox on Wasmtime and Tokio, with capability-based defaults.

Education

Polkadot Blockchain Academy 2023
UC Berkeley, California
Cryptography, Substrate, FRAME, Polkadot SDK, WebAssembly, decentralized networks.

Bachelor of Technology 2009 – 2013
GGSIPU, Delhi GPA: 7.5/10
Applied Mathematics, Data Structures, Algorithms, Distributed Computing, Cryptography.

Skills

- **Languages:** Rust (expert), Solidity, Go, TypeScript.
- **Protocol:** Substrate, FRAME, Polkadot SDK, governance pallet, WebAssembly, EVM.
- **On-chain exchange:** AMMs, stableswap, constant-product pools, liquidity accounting, swap execution.
- **Systems:** MVCC, io_uring, lock-free data structures, SIMD, Tokio, Wasmtime, performance engineering.
- **Data & infra:** PostgreSQL, Redis, AWS, Docker, CI/CD.